

CSA1280 – COMPUTER ARCHITECTURE

EXPERIMENT – 20

ASCENDING ORDER

AIM: To write and execute an 8085 assembly language program for ascending order and store result in memory.

ALGORITHM

1. Store the numbers in consecutive memory locations starting from **2050H**.
2. Load the count of numbers into register **C**.
3. Compare two consecutive numbers using the **CMP** instruction.
4. If the first number is greater than the second, interchange them.
5. Continue comparing all consecutive numbers.
6. Repeat the process until all numbers are arranged in ascending order.
7. Store the sorted numbers in the same memory locations.
8. Stop the program.

CODE:

```
MVI C,05H
```

```
LXI H,2050H
```

```
LOOP1: MOV A,M
```

```
    INX H
```

```
    CMP M
```

```
    JC NEXT
```

```
    JZ NEXT
```

```
    MOV B,M
```

```
    MOV M,A
```

```
    DCX H
```

```
    MOV M,B
```

```
    INX H
```

```
NEXT: DCR C
```

HLT

OUTPUT:

